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Advanced Data Mining - RS report

1. My decisions

1.1. Validation method

I chose **5-Fold Cross-Validation** instead of a hold-out set because it provides a better estimate. That's because in 5-fold cross-validation we basically create the hold-out set 5 times so it reduces the chances that we got "lucky" in the train/test split (which could happen in hold-out).

1.2. Custom MAE and RMSE

My `calculate_mae()` and `calculate_rmse()` functions were defined based on these formulas:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

They are the standard formulas so there is no point in explaining them here.

1.3. Algorithms

1.3.1. Naive

This is simply the average rating of the item across all users.

1.3.2. Collaborative filtering

I used the *KNNBasic* algorithm from the Surprise library because it is perfect for collaborative filtering. It also a memory-based algorithm - it memorizes the dataset and makes predictions based on similarities between users

As for configuration, I tested:

- different number of neighbors: `k` parameter
- different similarity measures: Cosine vs Mean Squared Difference (MSD) vs Pearson

MSD is the default measure in Surprise by the way.

2. Results

Algorithm	Parameters	MAE	RMSE
Naive (Item Average)	None	0.8174	1.0248
KNN Basic (Cosine)	k=40, sim=cosine	0.8044	1.0171
KNN Basic (MSD)	k=40, sim=MSD	0.7735	0.9789
KNN Basic (Pearson)	k=40, sim=pearson	0.8032	1.0120
KNN Basic (High Neighbors)	k=60, sim=MSD	0.7783	0.9840
KNN Basic (Low Neighbors)	k=10, sim=MSD	0.7793	0.9888

Table 1: Results of each configuration

3. Conclusions

These are my conclusions based on results in Table 1:

- **Collaborative Filtering vs Naive algorithms:** The Collaborative Filtering method is clearly better than naive algorithm (item average) - the best KNN got RMSE of 0.979 while naive got 1.024. The Naive method is not computationally expensive, but only calculates how an average user rates a movie without “personalization”.
- **Best Similarity Metric:** MSD (Mean Squared Difference) is the best metric for this dataset. It gave much lower errors (RMSE 0.979) than Cosine (1.017) and Pearson (1.012), which barely improved from the Naive method.
- **Neighborhood Size (k):** $k = 40$ achieved a little bit better results than $k = 60$ and $k = 10$ which means that there was too much noise when $k = 60$ (neighbors weren’t actually that similar) and there was not enough data (neighbors) when $k = 10$.

So, according to my experiments, the best model is **KNN Basic with MSD and k=40** for predicting ratings in this dataset.